

```

        """Drop the entire fleet and change the fleet's direction."""
        for alien in self.aliens.sprites():
            alien.rect.y += self.settings.fleet_drop_speed
    ❸ self.settings.fleet_direction *= -1

```

In `_check_fleet_edges()`, we loop through the fleet and call `check_edges()` on each alien ❶. If `check_edges()` returns `True`, we know an alien is at an edge and the whole fleet needs to change direction; so we call `_change_fleet_direction()` and break out of the loop ❷. In `_change_fleet_direction()`, we loop through all the aliens and drop each one using the setting `fleet_drop_speed` ❸; then we change the value of `fleet_direction` by multiplying its current value by `-1`. The line that changes the fleet's direction isn't part of the for loop. We want to change each alien's vertical position, but we only want to change the direction of the fleet once.

Here are the changes to `_update_aliens()`:

alien_invasion.py

```

def _update_aliens(self):
    """Check if the fleet is at an edge, then update positions."""
    self._check_fleet_edges()
    self.aliens.update()

```

We've modified the method by calling `_check_fleet_edges()` before updating each alien's position.

When you run the game now, the fleet should move back and forth between the edges of the screen and drop down every time it hits an edge. Now we can start shooting down aliens and watch for any aliens that hit the ship or reach the bottom of the screen.

TRY IT YOURSELF

13-3. Raindrops: Find an image of a raindrop and create a grid of raindrops. Make the raindrops fall toward the bottom of the screen until they disappear.

13-4. Steady Rain: Modify your code in Exercise 13-3 so when a row of raindrops disappears off the bottom of the screen, a new row appears at the top of the screen and begins to fall.

Shooting Aliens

We've built our ship and a fleet of aliens, but when the bullets reach the aliens, they simply pass through because we aren't checking for collisions. In game programming, *collisions* happen when game elements overlap. To make the bullets shoot down aliens, we'll use the function `sprite.groupcollide()` to look for collisions between members of two groups.